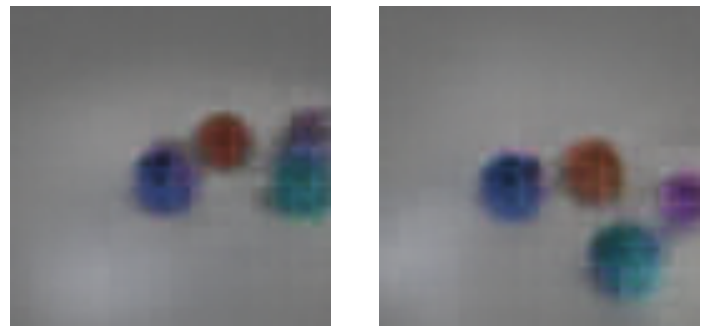


**Natural
language
utterance**

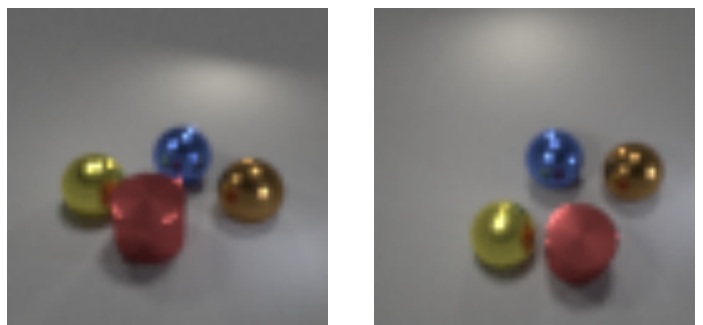
*“blue sphere to the left front of red
sphere to the left front of purple sphere
to the behind of cyan sphere”*

*“red cylinder to the right front of
yellow sphere to the left of brown
sphere to the right front of blue
sphere”*

**Neural
Renders**



**Blender
Renders**

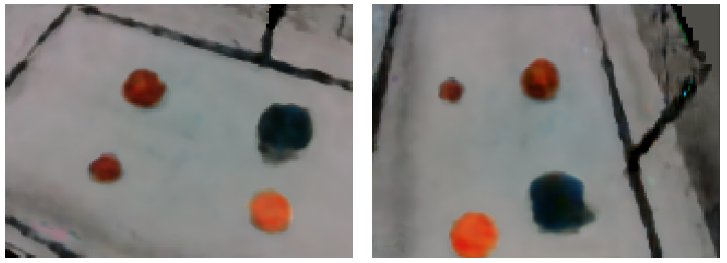


**Natural
language
utterance**

*“red apple to the left of orange to
the front of cup to the right of
pomegranate”*

*“toy truck to the left of yellow lemon to
the front of pomegranate to the right of
pear”*

**Neural
Renders**



View 1 (ref.)

View 2

View 1 (ref.)

View 2

(b) Given a language input, we visualize 2 different views generated from our model due to it’s 3-dimensional generation space